PRACTICAL CLOUD WORKBOOK

AWS Fundamentals

EC2 · S3 · IAM

A hands-on session for engineers moving from GCP and Azure into the AWS ecosystem.

LECTURE BY

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FORMAT

2-Hour Practical Lab

OVERVIEW

What is AWS?

Amazon Web Services is the world's most widely adopted cloud platform, offering on-demand compute, storage, and networking billed by usage.

200+

Services

Compute, storage, AI, and more

Global

Reach

Data centers across the world

Pay-go

Pricing

Only pay for what you use

FOUNDATION

AWS Global Infrastructure

AWS organizes its capacity into a hierarchy. Understanding it is the key to designing for resilience.

LEVEL 1

Region

A geographic area such as Mumbai or N. Virginia.

→

LEVEL 2

Availability Zone

One or more isolated data centers inside a Region.

→

LEVEL 3

Edge Location

Caching points that bring content closer to users.

CONTEXT

AWS vs Azure vs GCP

The same concepts, three vocabularies. Map what you already know.

AWS	Azure	GCP
Region / AZ	Region / Zone	Region / Zone
EC2	Virtual Machine	Compute Engine
S3	Blob Storage	Cloud Storage
IAM	Entra / RBAC	Cloud IAM

IDENTITY

IAM Overview

Identity and Access Management controls who can do what in your AWS account.

Users

An identity for a person or application with its own credentials.

Groups

A collection of users that share the same permissions.

Policies

JSON documents that define allowed or denied actions.

IDENTITY

IAM vs Azure RBAC vs GCP IAM

Every cloud answers the same question: who is allowed to do what.

AWS IAM

Users & Groups

JSON Policies

Roles for services

Azure RBAC

Entra identities

Role assignments

Scope-based access

GCP IAM

Members

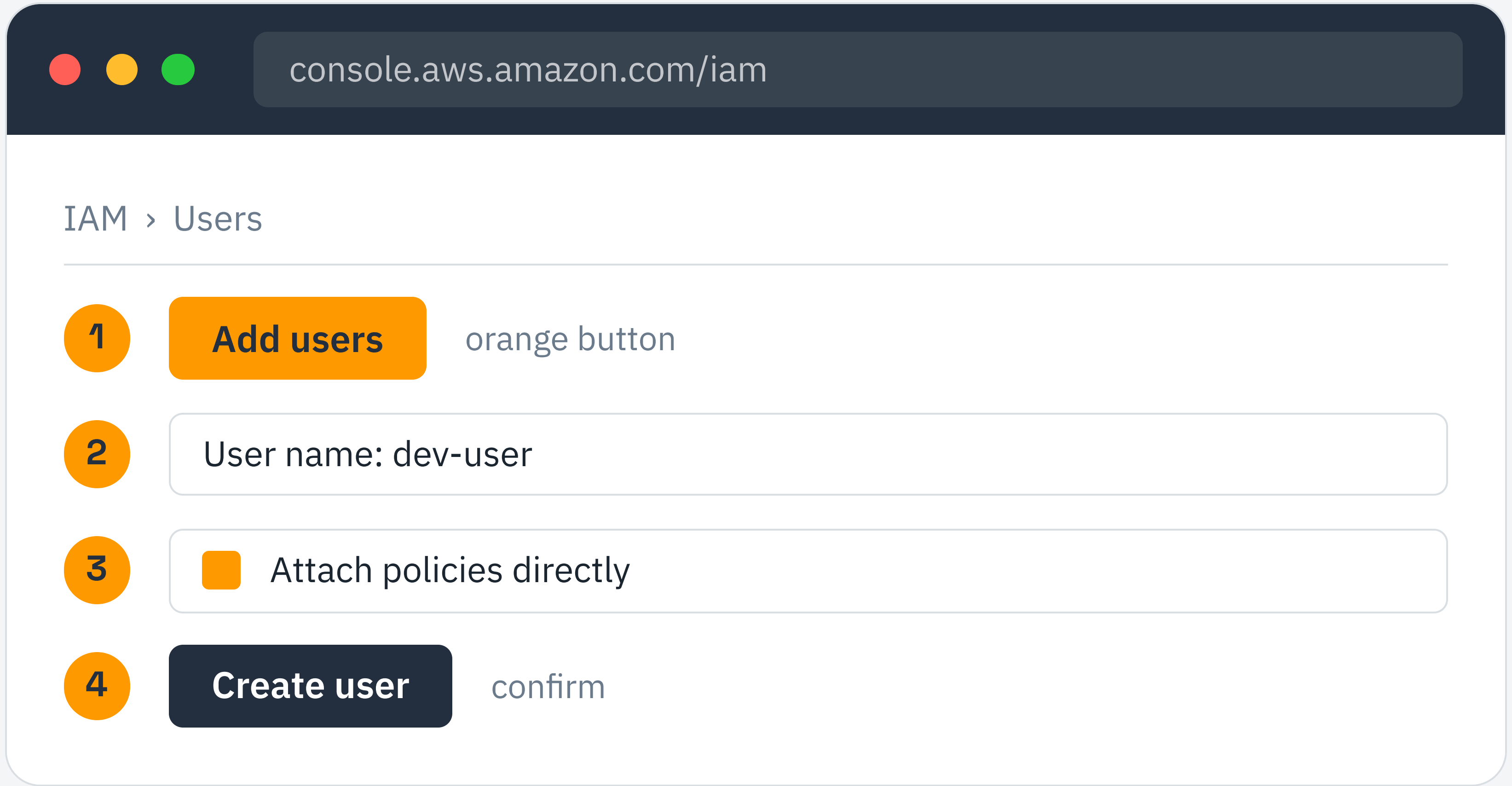
Predefined roles

Service accounts

WORKBOOK IDENTITY

Create an IAM User

1. Open the IAM console and choose Users, then Add users.
2. Enter a user name and enable console access.
3. Attach a policy or add the user to a group.
4. Review and create, then save the credentials.



SECTION 02 — COMPUTE

Compute Refresher

Compute is raw processing power on demand. In every cloud it means the same thing: virtual servers you can start, stop, and scale in minutes.

COMPUTE

What is EC2?

Elastic Compute Cloud provides resizable virtual servers in the cloud.
You choose the hardware, the OS, and pay only while it runs.

ELASTIC

Scale up or down on demand

Add capacity for a launch, remove it overnight.

FULL CONTROL

Root access to the machine

Install anything, configure the OS, manage networking.

COMPUTE

EC2 Architecture

An instance is the sum of its parts. Each piece is chosen at launch.

AMI

The OS and software template

Instance Type

CPU, memory, and network size

EBS Volume

Attached persistent storage

Security Group

Virtual firewall for traffic

KEY PAIR A public/private key that lets you securely SSH into the instance.

COMPUTE

Instance Types

Families are tuned for different workloads. Pick the shape that matches the job.

T **General Purpose**

Balanced for web apps and dev work

C **Compute Optimized**

High CPU for batch and gaming

R **Memory Optimized**

Large RAM for databases and caches

P **Accelerated**

GPUs for ML and rendering

COMPUTE

What is an AMI?

An Amazon Machine Image is a blueprint containing the OS, configuration, and software used to launch instances.

TEMPLATE
AMI

→

ACTION
Launch

→

RUNNING
EC2 Instance

One AMI can launch many identical instances — the foundation of repeatable, scalable deployments.

NETWORKING

Security Groups

A security group is a virtual firewall that controls inbound and outbound traffic to your instance.

Rules are allow-only and stateful — return traffic for an allowed request is automatically permitted.

NETWORKING

Security Groups vs NSG vs Firewall

Each cloud wraps your machine in a virtual firewall — the rules differ in name only.

AWS

Security Group

Stateful, allow-only rules per instance.

Azure

Network Security Group

Allow and deny rules with priorities.

GCP

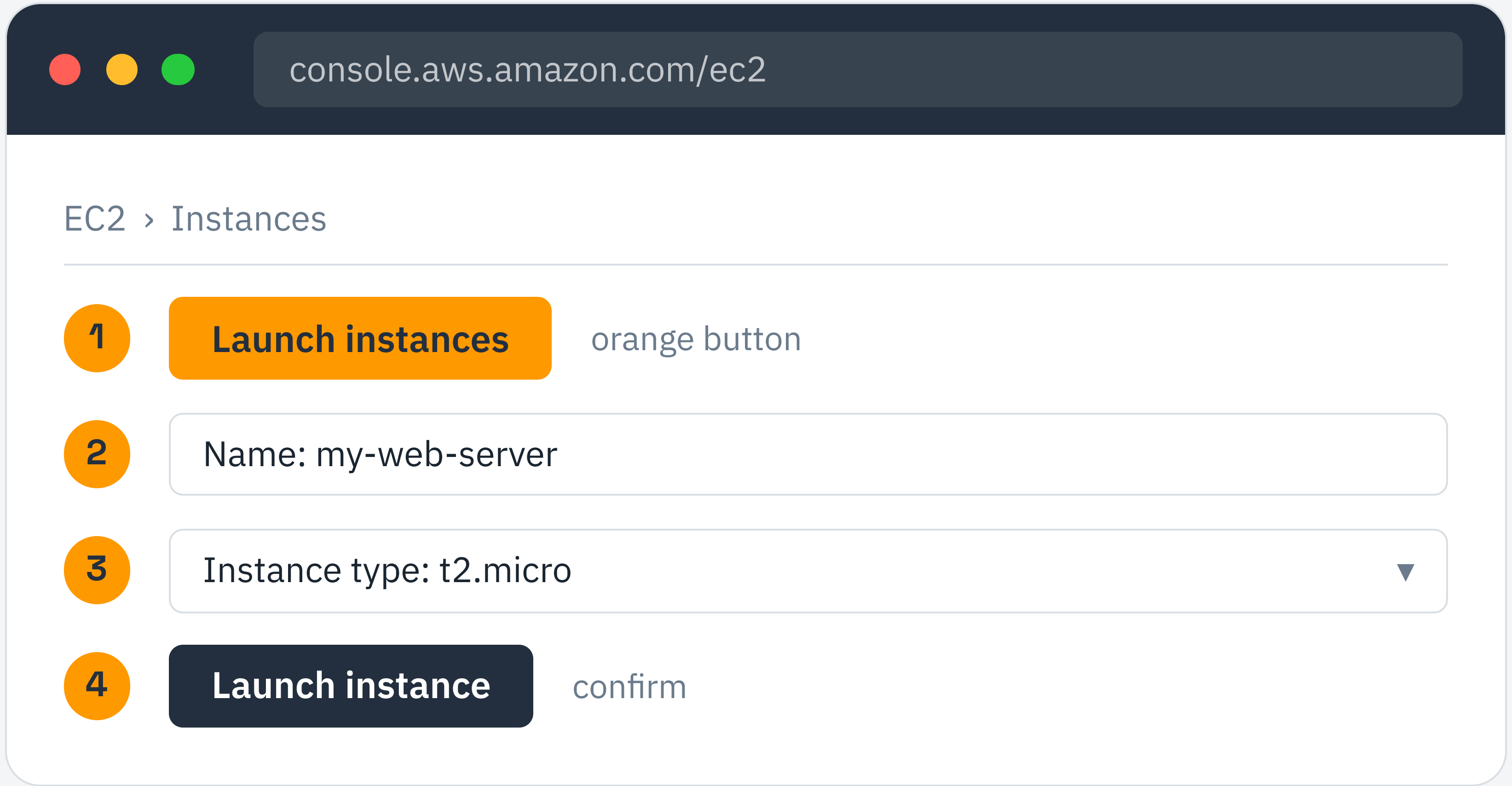
VPC Firewall Rules

Network-wide rules with target tags.

WORKBOOK COMPUTE

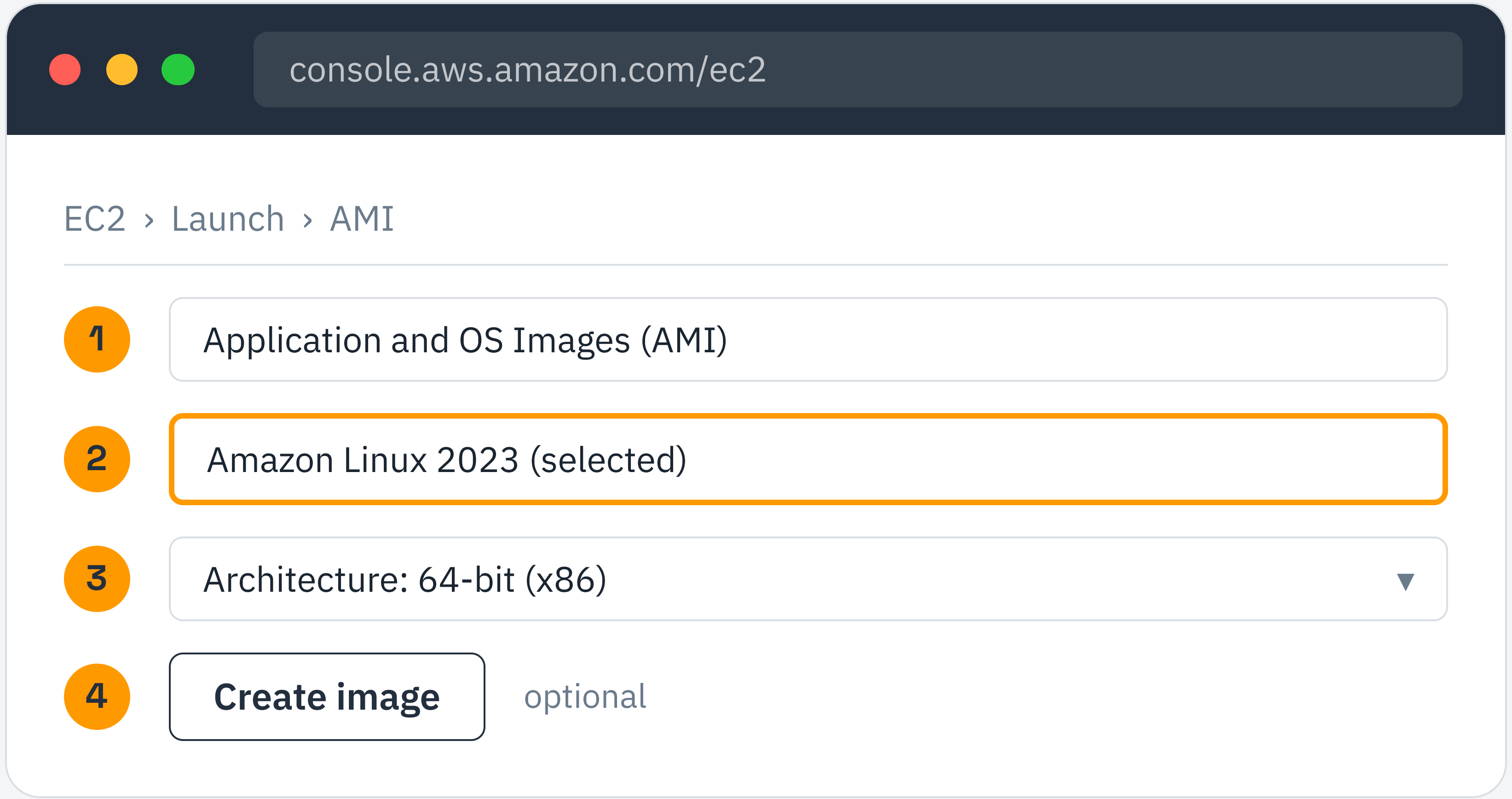
Launch an EC2 Instance

1. Open the EC2 console and choose Launch instance.
2. Name the instance and pick a region.
3. Select an instance type such as t2.micro.
4. Confirm settings and launch the instance.



Configure an AMI

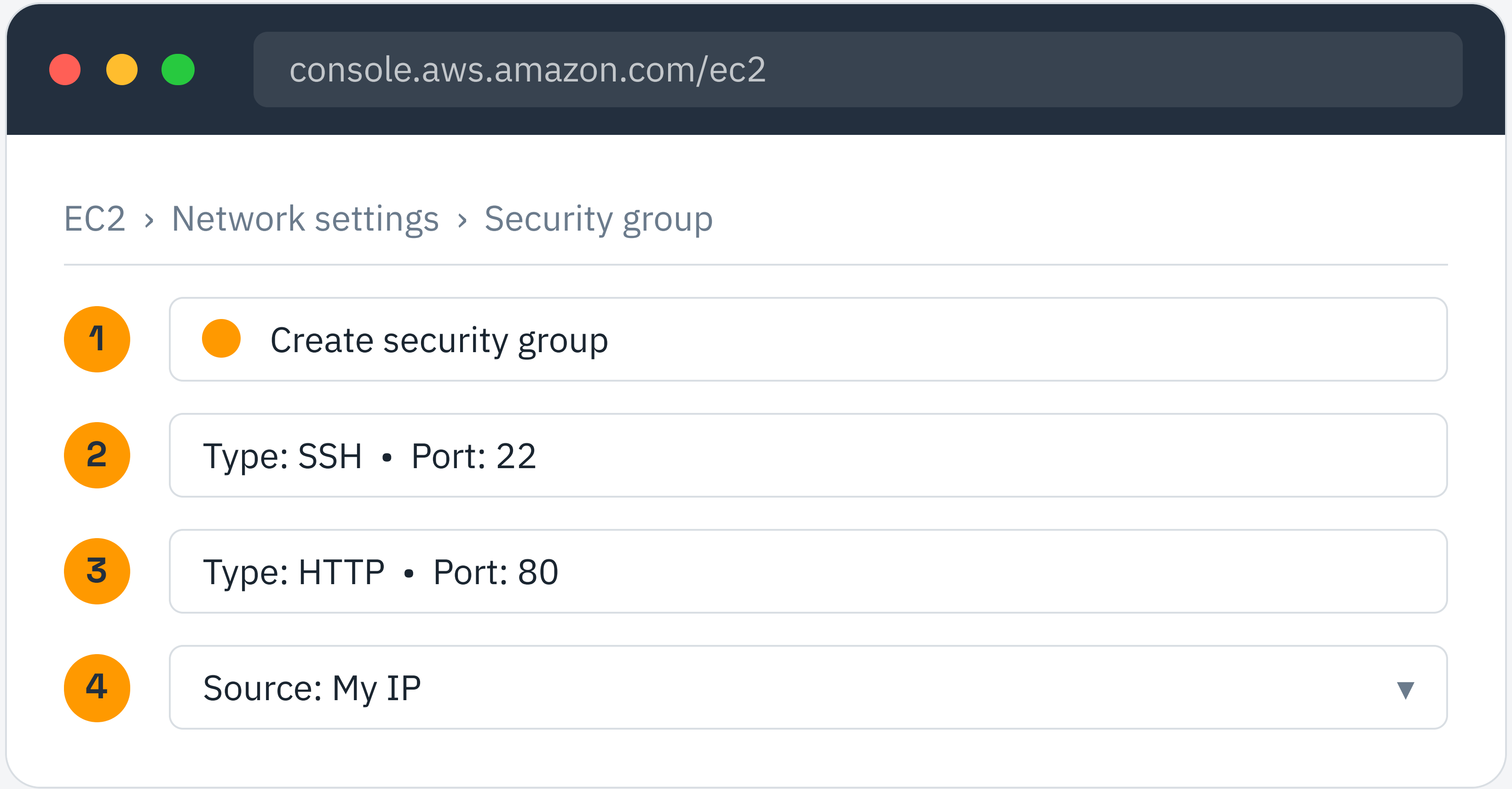
1. In the launch wizard, open Application and OS Images.
2. Choose a base AMI such as Amazon Linux or Ubuntu.
3. Confirm the architecture matches your instance type.
4. Optionally create a custom AMI from a running instance.



WORKBOOK NETWORKING

Configure a Security Group

1. In Network settings, create a new security group.
2. Add an inbound rule for SSH on port 22.
3. Add HTTP on port 80 if you run a web server.
4. Restrict the source to your IP for safer access.



WORKBOOK COMPUTE

Connect via SSH

1. Download your key pair file during instance launch.
2. Set file permissions with chmod 400 on the key.
3. Copy the instance public IP address.

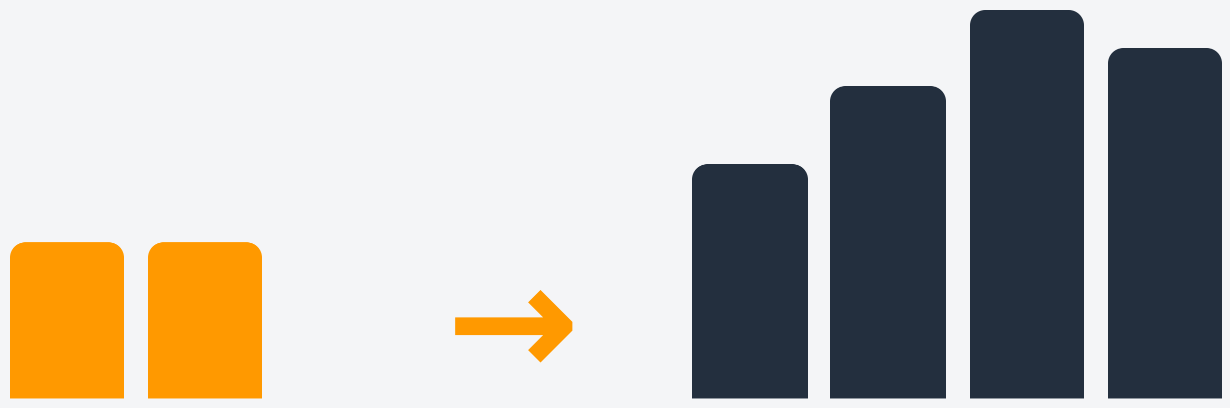
terminal

```
$ chmod 400 my-key.pem

$ ssh -i my-key.pem ec2-user@<public-ip>

Welcome to your instance
```

COMPUTE

Auto Scaling Concept

Auto Scaling automatically adjusts the number of instances to match demand, keeping performance steady and costs low.



Low traffic

2 instances

Peak traffic

Scaled out

COMPUTE

Why Auto Scaling Matters

It turns capacity planning from guesswork into an automatic response.

Availability

Replaces unhealthy instances automatically.

Cost Savings

Removes idle capacity when demand falls.

Performance

Keeps response times stable under load spikes.

SECTION 03 — STORAGE

Storage Refresher

Object storage holds files as self-contained objects with metadata, accessed over HTTP — ideal for backups, media, and static assets.

STORAGE

What is S3?

Simple Storage Service is object storage built to store and retrieve any amount of data with high durability.

11 nines

Durability

Data is replicated across facilities

Unlimited

Scale

No capacity to provision

HTTP

Access

Every object has a URL

STORAGE

Buckets & Objects

A bucket is a container. Objects are the files you store inside it.

BUCKET my-company-assets

logo.png image · 24 KB	**report.pdf** doc · 1.2 MB	**intro.mp4** video · 88 MB	**data.csv** file · 3 KB

Objects can be organized with key prefixes that look like folders, but S3 storage is flat underneath.

STORAGE

S3 vs Blob vs Cloud Storage

Object storage is universal — only the names and limits change.

AWS S3

- Buckets
- Objects
- Storage classes

Azure Blob

- Containers
- Blobs
- Access tiers

GCP Storage

- Buckets
- Objects
- Storage classes

WORKBOOK STORAGE

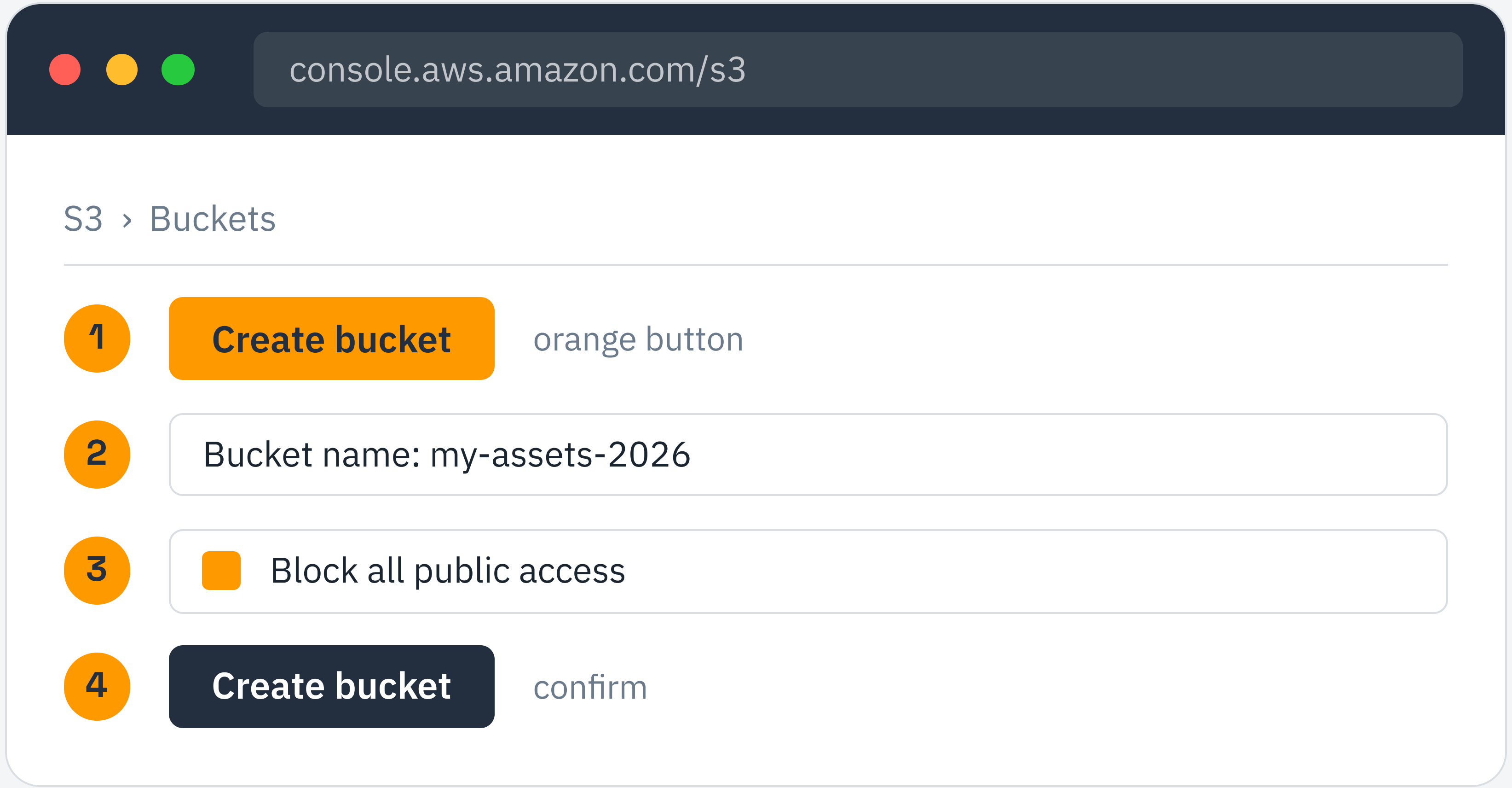
Create an S3 Bucket

1 Open the S3 console and choose Create bucket.

2 Enter a globally unique bucket name.

3 Keep Block Public Access on unless you need otherwise.

4 Review the settings and create the bucket.



WORKBOOK STORAGE

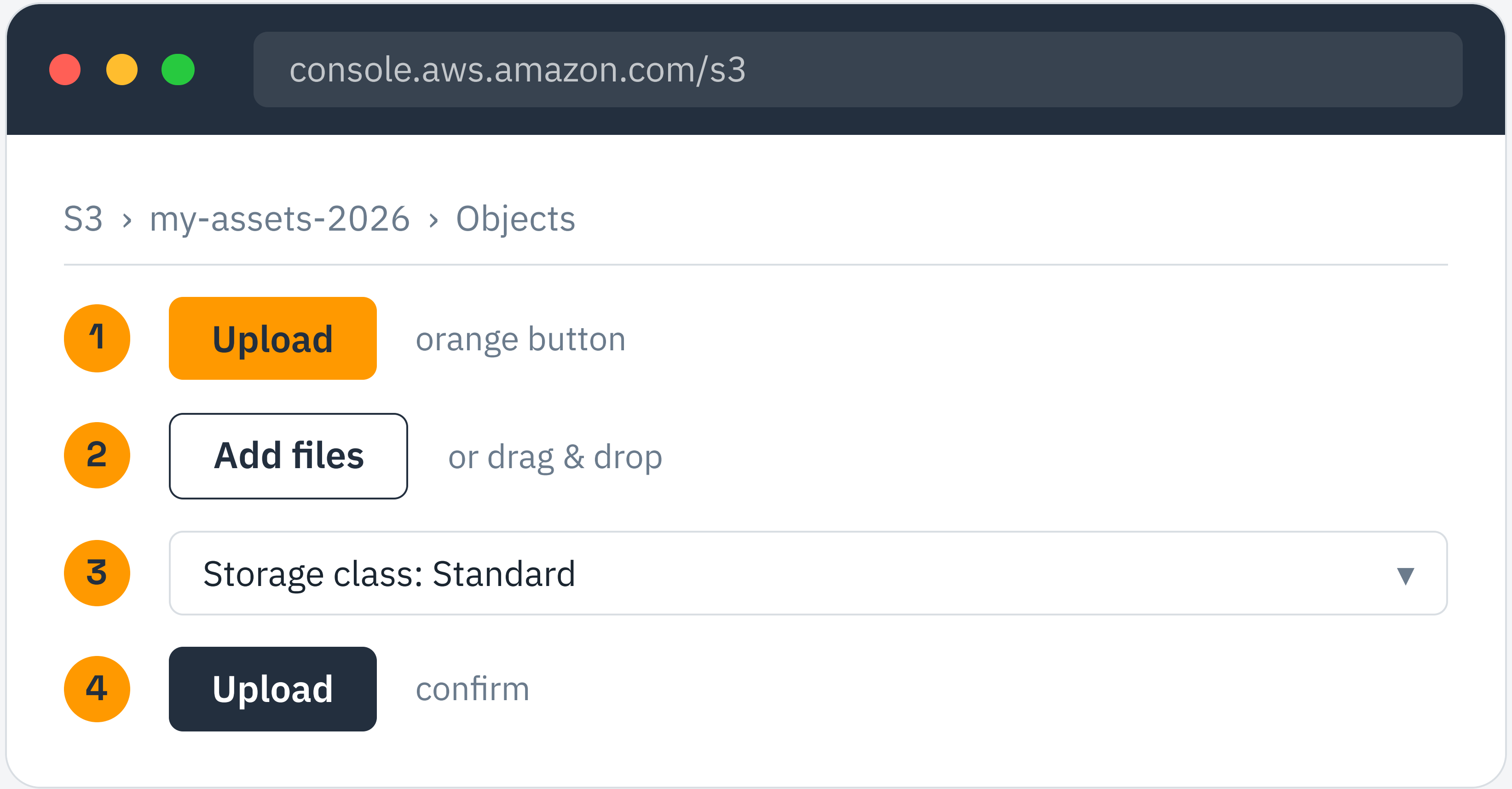
Upload Files to S3

1 Open your bucket and choose Upload.

2 Add files or drag them into the window.

3 Choose a storage class for the objects.

4 Confirm the upload and verify the objects appear.



Bucket Naming Rules

Bucket names are globally unique across all AWS accounts — no two buckets on Earth share a name.

Must be

- 3 to 63 characters long
- Lowercase letters and numbers
- Globally unique

Cannot be

- Formatted like an IP address
- Started or ended with a hyphen
- Using uppercase or spaces

MONITORING

Amazon CloudWatch

CloudWatch collects metrics, logs, and events so you can observe and react to what your infrastructure is doing.

Metrics

CPU, network, and disk over time.

Logs

Centralized application and system logs.

Alarms

Trigger actions when thresholds break.

MONITORING

CloudWatch vs Monitor vs Monitoring

Observability is built in to every cloud under a different brand.

AWS

CloudWatch

Metrics, logs, and alarms in one place.

Azure

Azure Monitor

Metrics and Log Analytics workspaces.

GCP

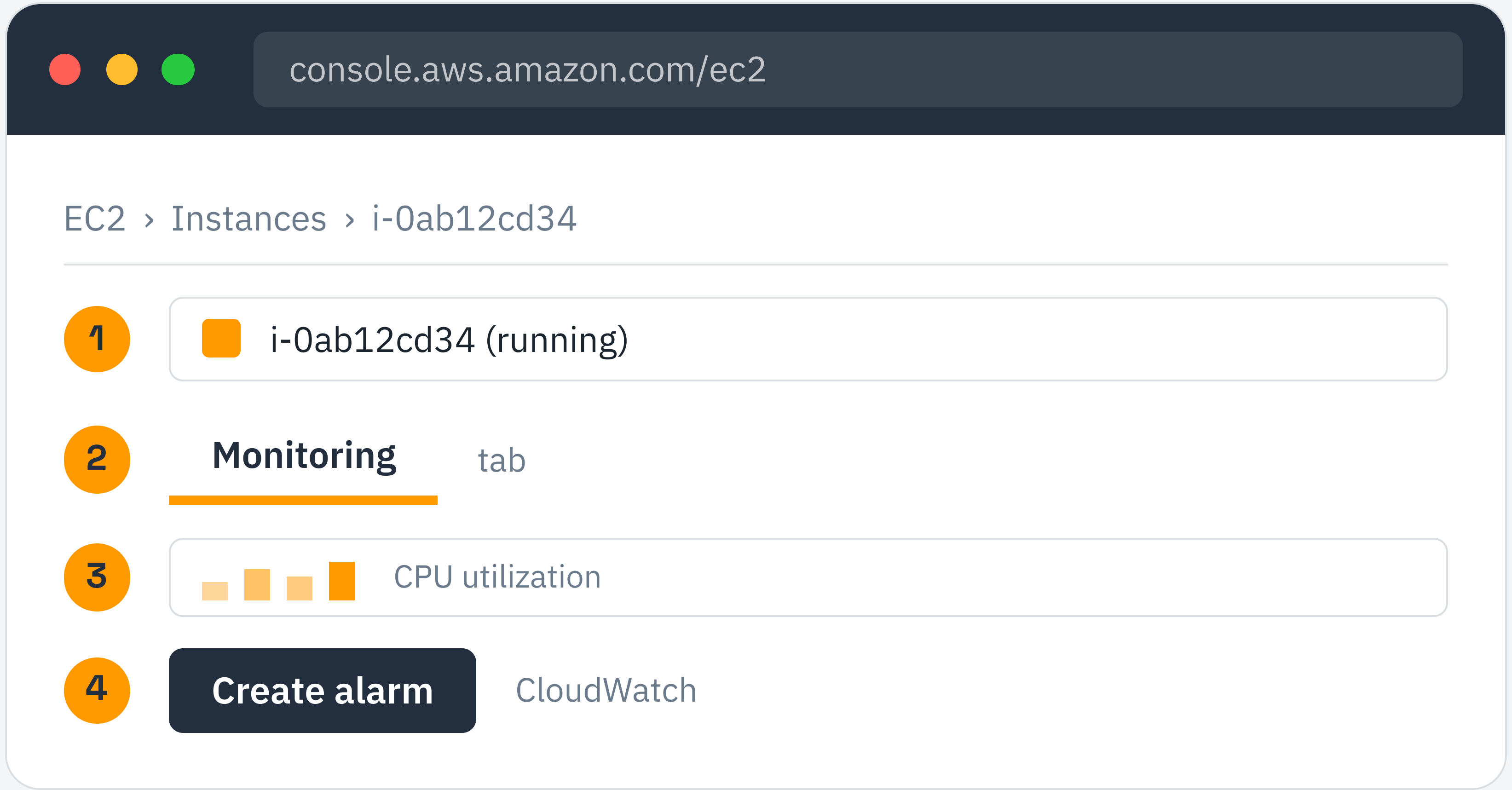
Cloud Monitoring

Part of the Operations suite.

WORKBOOK MONITORING

View EC2 Metrics

1. Select your instance in the EC2 console.
2. Open the Monitoring tab to see built-in graphs.
3. Review CPU utilization and network activity.
4. Create a CloudWatch alarm to get notified.



REVISION

MCQ Revision Sheet

1. Which service provides virtual servers?

A) S3   B) EC2   C) IAM   D) CloudWatch

2. Where are S3 files stored?

A) Buckets   B) Volumes   C) Groups   D) Zones

3. What controls instance traffic?

A) AMI   B) IAM   C) Security Group   D) Bucket

4. Which service manages permissions?

A) EC2   B) IAM   C) S3   D) AMI

ANSWERS   1-B   2-A   3-C   4-B

REVISION

AWS Cheat Sheet

EC2

Virtual servers — launch from an AMI, sized by instance type.

S3

Object storage — globally unique buckets holding objects.

IAM

Access control — users, groups, and JSON policies.

CloudWatch

Monitoring — metrics, logs, and alarms.

CHALLENGE APPLY IT

Hands-On Challenge

Combine today's services into one working deployment.

1. Create an IAM user with EC2 and S3 permissions.
2. Launch an EC2 instance and connect over SSH.
3. Create an S3 bucket and upload a sample file.
4. Open CloudWatch and watch your instance metrics.

REVISION

Interview Questions

Q1 What is the difference between an AMI and an instance?

Q2 How does a security group differ from a network ACL?

Q3 Why must S3 bucket names be globally unique?

Q4 When would you use an IAM role instead of a user?

Q5 How does Auto Scaling decide when to add instances?

WRAP UP

You Now Know AWS Core

EC2 Compute

S3 Storage

IAM Identity

CloudWatch

Keep building — the best way to learn the cloud is to deploy in it.

PRESENTED BY

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